

GRAPHICS on the FPS M64

INTRODUCTION

Computer graphics techniques have matured to form a rich and powerful means of communicating information. As algorithms have evolved, they require increasing amounts of computer time. Modern scientific computers have evolved in parallel to satisfy the requirements of complex scene generation programs. The FPS M64 Series is a networked computer system, ideal for rendering complex 3-D images, generating interpretive views of scientific experiments, and research into advanced graphics algorithms.

The architecture and software is compatible over the entire FPS M64 range, allowing code developed on the small, low cost, entry level devices to be moved to higher performance machines at a later date, or sent to a central facility for accelerated turn-around. The following shows the 64 bit floating point performance range of the FPS M64 Series:

M64/20	6 Mflops
M64/30	12 Mflops
M64/40	11 Mflops
M64/50	20 Mflops
M64/60	38 Mflops

The image rendering phase of generating 3-D animated sequences needs extremely powerful computer systems if they are to be effective in a short production schedule. Advanced scientific computer architectures connected together on local area networks can provide the high level of computer power needed for the animator. This paper shows the characteristics of the FPS M64 Series that give best advantage to complex rendering algorithms. Commercial software packages available for direct execution on the FPS M64 means no development effort is required for the first production runs.

RENDERING

Production animation systems have many diverse components, usually organised by the type of tool required at each stage of the production cycle. After the story board and artistic design, comes model building and motion dynamics. These can be previewed on fast graphics workstations, or by generating line tests on film or video. After most of the scene has been designed, the computer system is finally set to work generating high-resolution, full colour images for each frame in the animation sequence. The generation of these images is called the 'rendering' step. This single task can consume as much as 95% of the entire computer system, and may therefore be the limiting factor for turning out the film segment within the production deadlines. The FPS M64 network compute server can provide the dramatic boost in performance just where it is needed: during the final rendering!

After the geometric form of the scene has been determined, each pixel in the image must be assigned a colour appropriate to the light contribution to that point in the scene. Lighting models are used to determine the colour and intensity from the geometric information. One of the popular lighting models, and the most

easily implemented, is PHONG shading. The PHONG shading model combines ambient, diffuse, and specular lighting components to derive the final shade at a surface point due to one light source. This vectorises well using high speed vector math library routines supplied with each system.

SOFTWARE DESIGN

Modern graphics software systems developed in 'C' can run directly on the FPS M64 network stations, by using the resident compiler and linker. An optimising FORTRAN-77 compiler is also available, for a broad range of scientific code and package development. The program development tools are resident on the FPS M64, thus source code can be sent to the system together with data files. Executable modules are held on the local disk system, so rendering programs can be started with very low overheads. The FPS software has state-of-the-art optimisation techniques for using the full performance of resources on the FPS M64.

The FPS M64 is accessed either by users on the local workstation to which the FPS M64 is attached, or by remote access from other workstations on the network (eg. Ethernet).

PACKAGES

Several powerful rendering programs currently are running on the FPS M64 Series. At SIA, a solid modelling Ray Tracing program is available for scene generation, also the MOVIE.BYU polygon rendering system from Brigham Young University. Versatec Rasterisation software runs directly on the FPS M64 series, eliminating the need for special purpose rasterisation processors. The CPS1 Contouring package from Radian performs 3-D spatial contours for the Oil Reservoir contour plotting.

The software used to generate displays on Versatec plotters has been adapted for FPS M64 Series Minisupercomputers configured with VAX front-ends running VMS. Versatec Color Random Version 1.0 is a subroutine library that is called by the user's Fortran program to convert data files into graphic displays

Software packages such as Versatec Color Random translate output file data into graphics ranging from simple line plots to full colour, two-dimensional plots, area fills with a variety of patterns, logarithmic grids, and titles and borders.

GRAPHICS RESEARCH

Many research labs and universities are using the FPS M64 Series for graphics research. Their work includes recently published work in iterated function theory (fractals), ray tracing, terrain simulation, and radiosity methods.

RAY TRACING RESEARCH

Ray Tracing is, perhaps, the simplest and most accurate method of synthesizing a complex three dimensional image. In physical terms, the algorithm retraces the paths of light rays entering the eye, taking into account the rays' reflections and refractions, and the effect of light scattering surfaces. Unfortunately, ray tracing is extremely computationally intensive. The number of calculations can become large very quickly with increases in the number of objects in the scene, the depth of the ray tree, and the

image resolution.

In ray tracing algorithms, at least 85% of the execution time is spent in computing ray and object geometric intersections. The type of intersection calculation depends on the complexity of the object description, but is usually some type of polynomial root finder. For quadric or polygonal objects, the ray and object intersection test can be vectorised using the FPS vector math library. ■

Ray tracing is an ideal test bed for experimenting with new types of surface properties, optical effects, and the physical behavior of light and matter. Statistical sampling of ray paths can simulate diffuse reflection, roughened surface scattering, area light sources, motion blur, and depth of field effects.

RADIOSTY RESEARCH

Radiosity is a major new method for simulating complex 3-dimensional scenes, accounting for subtle effects such as the diffusion of bounced light into shadows, blending of the colour of one object onto its neighbours, and caustics illumination. This method draws upon the physics of thermal heat transfer, and was first published by Cindy Goral, formerly of FPS Software Engineering.

The FPS M64/60 was instrumental in this research, which requires 8 days of CPU time for a single scene generation run.

As part of the Cornell Centre for production supercomputing, the FPS M64/60 is located in the Computer Graphics Laboratory and used for State-of-the-art graphics research as well as other scientific research.

WHY 64 BITS ?

The state-of-the-art rendering algorithms are nearly as complex as the scenes they render, and techniques for improving computational efficiency are still an active research topic. The complexity of the scene has a direct effect on the precision requirements of the rendering algorithm.

It is interesting to note that in most C compiler implementations, C programs compute floating point operations in 64 bits, regardless of the floating point data types referenced in the source code. Hence many graphics programs are spending CPU resources that they may not be fully using.

The perspective projection, coupled with a large number of polygons, introduces the requirement for extended precision to test visibility.

An additional problem requiring extended precision is the calculation of surface normals for specular reflections. Highlights or bright spot reflections from object surfaces are computed by a separate term in most lighting models. The highlight values are a function of polygon surface normal angle, light source direction normal, and viewer direction angle. These three direction vectors are compared, and the cosine of the difference angle raised to a high power. The typical formulation

is numerically similar to

intensity = (COS (alpha))**25,
where alpha = angular difference of surface reflection
direction, and viewer direction.

If there is a numerical error in the surface normal calculations, the highlights tend to jitter or jump around when animated.

In advanced scene rendering, complex lighting effects such as the reflections of one object inside another, refraction through semi-transparent objects, and shadows from multiple light sources depend even more strongly on accurate reflection angles.

CONCLUSION

Large amounts of memory and extended arithmetic precision are required to produce images where both realism and high scene complexity converge. The areas of perspective depth comparisons, anti-aliasing, and polynomial evaluation for ray tracing have very large 64-bit computational requirements.

The FPS M64 Series of Scientific Computers are well balanced to solve these complex rendering problems and represent powerful tools for digital scene simulation on an affordable scale.

The FPS M64 Series includes office environment versions, for connection to workstations either directly or through Ethernet local area networks.

For further information on topics mentioned in this paper contact Ian Curington or Tim Allen at Floating Point Systems UK, Ltd., Apex House, London Road, Bracknell Berkshire, RG12 2TE, (0344) 56921.

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